Post-Intubation / BIAD Management

AR-08 | Airway & Respiratory

Quick flow

01 ETT / BIAD Placed [EMT / AEMT / Paramedic]

Confirm success with continuous waveform $EtCO_2$, bilateral exam, SpO_2, tube depth, and securement.

02 Secure + Ventilate [EMT / AEMT / Paramedic]

Manual stabilization during movement • appropriate rate • visible chest rise • $EtCO_2$-guided reassessment.

03 Monitor Continuously [EMT / AEMT / Paramedic]

Waveform $EtCO_2$ • SpO_2 • cardiac monitor • tube depth • breath sounds • frequent vital signs.

04 Pain / Anxiety / Movement? [Paramedic]

Paramedic: treat pain first, then add sedation when needed. Pediatric medication requires pediatric pathway/online order.

05 Sudden Deterioration? [EMT / AEMT / Paramedic]

Evaluate DOPE; disconnect ventilator and manually ventilate if cause is not immediately clear.

06 Suspected TBI? [EMT / AEMT / Paramedic]

SpO_2 ≥94%; urgently correct <90% or inadequate ventilation. Initial/routine assisted $EtCO_2$ 35-45 mmHg. No routine hyperventilation. Clear active/imminent herniation only: brief $EtCO_2$ 30-35 with capnography, avoid <30, and expedite transport. Open TB-05.

07 Transport + Reconfirm [EMT / AEMT / Paramedic]

Recheck after every move, document airway status/medications, and notify early.

Overview

- After every ETT or BIAD placement, continuously confirm airway position, oxygenation, ventilation, securement, analgesia, and sedation.
- Treat pain first; paralysis does not provide analgesia or sedation. Newly born patients remain under AO-02.

Provider-Level Actions

Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT

- Manually stabilize and secure the device, ventilate at the appropriate rate, monitor SpO_2, and reassess after every move.
- Immediately report loss of waveform, resistance, reduced breath sounds, hypoxia, or changing tube depth.

AEMT

- Perform all EMT care plus waveform capnography, IV/IO access, authorized ventilation support, suctioning, and assistance with DOPE management.
- This protocol does not grant AEMT authority for post-intubation analgesics, sedatives, or paralytics.

Paramedic

- Perform all prior care plus pain-first adult analgesia/sedation, mechanical ventilation, and rescue management for DOPE deterioration.
- For pediatric patients, use the length-based pediatric medication system and direct online Medical Control or receiving-facility orders for post-airway analgesia/sedation; do not apply adult doses.

- Use paralysis only as a last resort after adequate analgesia and sedation. If movement remains unsafe, use paramedic judgment with continuous sedation and Medical Control or an established interfacility order.

Indications

- Any patient after successful endotracheal intubation or BIAD placement, excluding neonatal resuscitation immediately after birth.

Treatment steps

- Confirm ETT/BIAD placement with continuous waveform $EtCO_2$, bilateral chest examination, SpO_2, tube depth, and securement. Recheck all confirmation points after every transfer or move.
- Ventilate adults at approximately 10-12 breaths/min and pediatric patients at 20-30 breaths/min (1 breath every 2-3 seconds), with visible chest rise. Use $EtCO_2$ and clinical response to avoid hyperventilation.
- Avoid hypoxemia. After ROSC, target adult SpO_2 90-98% and pediatric SpO_2 94-99% once reliable monitoring is available.
- For sudden deterioration, use DOPE: displacement, obstruction, pneumothorax, equipment failure. Disconnect from the ventilator and manually ventilate if the cause is not immediately clear.
- Suction when indicated, elevate the head 10-20 degrees when feasible, and reassess breath sounds, tube depth, and waveform after suctioning or movement.
- Treat pain first, then add sedation. Do not stack fentanyl and morphine; select one analgesic strategy.
- Document airway confirmation, tube depth, waveform $EtCO_2$, oxygenation, ventilation rate/settings, analgesia/sedation, reassessments, and all movement-related rechecks.
- For suspected traumatic brain injury, continuously monitor oxygenation and ventilation. Aim for SpO_2 at least 94%; urgently correct SpO_2 below 90% or inadequate ventilation. The condition-specific TBI target takes priority over a lower general airway or post-ROSC oxygenation target.
- For suspected traumatic brain injury requiring assisted ventilation, use an initial/routine $EtCO_2$ target of 35-45 mmHg. Avoid routine hyperventilation. Only for clear active or imminent cerebral herniation, use brief controlled ventilation targeting $EtCO_2$ 30-35 mmHg with continuous waveform capnography while expediting definitive care; avoid overshooting below 30 mmHg.
- Prevent hypotension in suspected traumatic brain injury; do not use permissive hypotension. For adults, target SBP at least 110 mmHg, including age 70, with frequent reassessment and after treatment. Pediatric TBI blood-pressure targets require age-specific direction from Medical Control until the separately approved pediatric TBI target table is published; do not substitute adult targets or generic pediatric shock thresholds.

Medications

Ketamine - adult

1 mg/kg IV/IO

- Adult alternative for post-airway sedation when clinically appropriate. Pediatric post-airway medication requires the pediatric system and direct online Medical Control or receiving-facility order.

Fentanyl - adult

25-50 mcg IV/IO; repeat cautiously as needed, maximum 100 mcg

- Treat pain first. Do not stack with morphine.

Morphine - adult alternative

4 mg IV/IO; repeat 2 mg every 5 minutes as needed, maximum 10 mg

- Select one analgesic strategy; do not stack with fentanyl.

Midazolam - adult

2-5 mg IV/IO; repeat cautiously as needed

- Use after analgesia when sedation is needed.

Warnings / precautions

- Waveform capnography is mandatory after ETT and after BIAD placement once available.
- A paralyzed patient may be awake and in pain without visible movement; maintain analgesia and sedation.
- Loss of waveform or sudden deterioration requires immediate DOPE evaluation; never delay manual ventilation for ventilator troubleshooting.
- Propofol may not be initiated or bolused by field Paramedics; an already ordered interfacility infusion may continue when appropriate.
- Do not apply adult post-airway medication doses to pediatric patients.
- This protocol does not authorize routine repeat paralysis. Pediatric post-airway medication requires the pediatric pathway and a direct online Medical Control or receiving-facility order.

Clinical pearls

- Elevate the head 10-20 degrees when possible to reduce aspiration risk.
- For sudden deterioration, disconnect the ventilator, manually ventilate, and evaluate DOPE.
- A secure airway requires continuous confirmation, not a single post-placement check.

Related pathways / contacts

AR-03 Drug-Assisted Airway - Open for the approved DAI preparation, induction, and post-airway sequence.

AR-05 Pediatric Airway - Open for pediatric airway support and the length-based medication system.

AR-09 Ventilator Emergencies - Open for ventilator alarm, equipment failure, or unresolved ventilator deterioration.

AR-11 Mechanical Ventilation, Adult - Open for adult ventilator setup and management.

TB-05 Head Trauma - Open the shared TBI oxygenation, ventilation, herniation-rescue, and perfusion targets.

Source and references

Claiborne County EMS approved protocol manual. | Claiborne County EMS formulary, linked protocol pathways, and agency release guidance.

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